

This is unreleased documentation for Serverpod **Next** version.

For up-to-date documentation, see the [latest version](#) (4.0.0).

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MODEL REFERENCE

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## Model reference

Every keyword available in a Serverpod model file, and whether it applies to a `class`, `exception`, or `enum`. For how each one works, see [Working with models](#).

| Keyword                        | Note                                                                                                                                 | class | exception | enum |
|--------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|-------|-----------|------|
| <code>values</code>            | A special key for enums with a list of all enum values.                                                                              |       |           | ✓    |
| <code>serialized</code>        | Sets the mode enums are serialized in.                                                                                               |       |           | ✓    |
| <code>properties</code>        | Defines custom properties for enhanced enums.                                                                                        |       |           | ✓    |
| <code>immutable</code>         | Boolean flag to generate an immutable class with final fields, equals operator, and hashCode.                                        | ✓     |           |      |
| <code>serverOnly</code>        | Boolean flag if code generator only should create the code for the server.                                                           | ✓     | ✓         | ✓    |
| <code>table</code>             | A name for the database table, enables generation of database code.                                                                  | ✓     |           |      |
| <code>database</code>          | Where a table lives: <code>server</code> (default), <code>client</code> , <code>all</code> , or the experimental <code>sync</code> . | ✓     |           |      |
| <code>managed Migration</code> | A boolean flag to opt out of the database migration system.                                                                          | ✓     |           |      |



| Keyword                      | Note                                                                                                                                                              | class | exception | enum |
|------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------|------|
| <b>serializationDataType</b> | Stores serializable fields as <code>json</code> or <code>jsonb</code> . Valid on classes and on individual fields.                                                | ✓     |           |      |
| <b>fields</b>                | All fields in the generated class should be listed here.                                                                                                          | ✓     | ✓         |      |
| <b>type (fields)</b>         | Denotes the data type for a field.                                                                                                                                | ✓     | ✓         |      |
| <b>required</b>              | Makes the field required. This keyword can only be used for <b>nullable</b> fields.                                                                               | ✓     | ✓         |      |
| <b>tail</b>                  | Places the field after all normal fields of an inheritance hierarchy. Child tail fields come before parent tail fields. Not allowed on the <code>id</code> field. | ✓     | ✓         |      |
| <b>scope</b>                 | Denotes the scope for a field.                                                                                                                                    | ✓     |           |      |
| <b>jsonKey</b>               | Sets a custom key name for JSON serialization and deserialization.                                                                                                | ✓     |           |      |
| <b>column</b>                | Overrides the database column name for a field.                                                                                                                   | ✓     |           |      |
| <b>persist</b>               | A boolean flag if the data should be stored in the database or not can be negated with <code>!persist</code>                                                      | ✓     |           |      |
| <b>relation</b>              | Sets a relation between model files, requires a table name to be set.                                                                                             | ✓     |           |      |
| <b>name</b>                  | Give a name to a relation to pair them.                                                                                                                           | ✓     |           |      |
| <b>parent</b>                | Sets the parent table on a relation.                                                                                                                              | ✓     |           |      |
| <b>field</b>                 | Names the foreign key field of a relation. Generated if the field is not declared on the model.                                                                   | ✓     |           |      |
| <b>fk</b>                    | Boolean flag marking the side of a bidirectional relation that holds the foreign key, keeping the generated field name.                                           | ✓     |           |      |

| Keyword                         | Note                                                                                                                                        | class | exception | enum |
|---------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------|------|
|                                 | Mutually exclusive with <b>field</b> .                                                                                                      |       |           |      |
| <b>onUpdate</b>                 | Set the referential actions when updating data in the database.                                                                             | ✓     |           |      |
| <b>onDelete</b>                 | Set the referential actions when deleting data in the database.                                                                             | ✓     |           |      |
| <b>optional</b>                 | A boolean flag to make a relation optional.                                                                                                 | ✓     |           |      |
| <b>deferrable</b>               | Make the foreign key constraint deferrable, checked after each statement unless a transaction defers it.                                    | ✓     |           |      |
| <b>deferred</b>                 | Make the foreign key constraint deferrable and checked at commit. Mutually exclusive with <b>deferrable</b> .                               | ✓     |           |      |
| <b>indexes</b>                  | Create indexes on your fields / columns.                                                                                                    | ✓     |           |      |
| <b>fields (index)</b>           | List the fields to create the indexes on.                                                                                                   | ✓     |           |      |
| <b>type (index)</b>             | The type of index to create.                                                                                                                | ✓     |           |      |
| <b>parameters (index)</b>       | Parameters for specialized index types like HNSW and IVFFLAT vector indexes.                                                                | ✓     |           |      |
| <b>distanceFunction (index)</b> | Distance function for vector indexes. Allowed values depend on the field's vector type and index type, see <a href="#">vector indexes</a> . | ✓     |           |      |
| <b>operatorClass (index)</b>    | Operator class for GIN indexes.                                                                                                             | ✓     |           |      |
| <b>unique</b>                   | Creates a unique index. Also valid directly on a field, optionally scoped with <code>unique(per=...)</code> .                               | ✓     |           |      |
| <b>nulls_distinct (index)</b>   | Whether null values collide in a unique index. Only allowed on unique indexes.                                                              | ✓     |           |      |

| Keyword               | Note                                                                                                                                                                              | class | exception | enum |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------|-----------|------|
| <b>default</b>        | Sets the default value for both the model and the database. This keyword cannot be used with <b>relation</b> . On an enum, sets the <a href="#">fallback for unknown values</a> . | ✓     | ✓         | ✓    |
| <b>defaultModel</b>   | Sets the default value for the model side. This keyword cannot be used with <b>relation</b> .                                                                                     | ✓     | ✓         |      |
| <b>defaultPersist</b> | Sets the default value for the database side. This keyword cannot be used with <b>relation</b> and <b>!persist</b> .                                                              | ✓     |           |      |
| <b>extends</b>        | Specifies a parent class or exception of the same kind to inherit from.                                                                                                           | ✓     | ✓         |      |
| <b>sealed</b>         | Boolean flag to create a sealed class or exception hierarchy, enabling exhaustive type checking.                                                                                  | ✓     | ✓         |      |

## Related

- [Working with models](#): how each keyword works, with examples.
- [Field types](#): YAML types mapped to Dart, Postgres, SQLite, and JSON.
- [Dynamic fields](#): store and transfer values whose type varies at runtime.
- [Serializable exceptions](#): define, inherit, throw, and catch application errors.
- [Database tables](#): the database-specific keywords (`table`, `relation`, `persist`, and indexing).